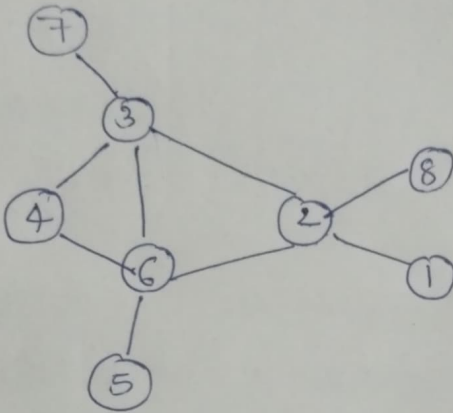


①



a) geodesic distance

b) eccentricity

c) average shortest path

d) radius

e) diameter

f) clustering coefficient

g) degree centrality

h) Eccentricity centrality

i) closeness centrality

j) betweenness centrality

a)

	1	2	3	4	5	6	7	8	
1	0	1	2	3	3	2	3	2	=16
2	1	0	1	2	2	1	2	1	9
3	2	1	0	1	2	1	1	2	7
4	3	2	1	0	2	2	2	3	8
5	3	2	2	2	0	1	3	3	7
6	2	1	1	1	1	0	2	2	4
7	3	2	1	2	3	2	0	3	3
8	2	1	2	3	3	2	3	0	

b) eccentricity

$$e(1) = 3$$

$$e(2) = 2$$

$$e(3) = 2$$

$$e(4) = 3$$

$$e(5) = 3$$

$$e(6) = 2$$

$$e(7) = 3$$

$$e(8) = 3$$

c) avg shortest path

$$= \frac{54}{28} = 1.92$$

d) radius = 2

e) diameter = 3

g)

1	1
2	4
3	4
4	2
5	1
6	4
7	1
8	1

h)

1	0.33
2	0.5
3	0.5
4	0.33
5	0.33
6	0.5
7	0.33
8	0.33

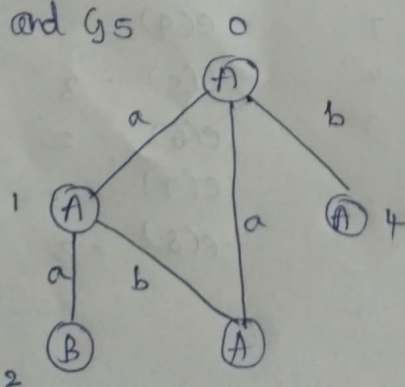
i) CCV) - $\frac{1}{\sum d(u,v)}$

1	0.077
2	0.125
3	0.125
4	0.088
5	0.077
6	0.125
7	0.077
8	0.077

(j) Betweenness

2) G₂ and G₅

3)



(0, 1, A, a, A)

(1, 2, A, a, B)

(1, 3, A, b, A)

(0, 3, A, a, A)

(0, 4, A, b, A)

DFS order

1. $A \xrightarrow{a} A$

2. $A \xrightarrow{a} B$

3. Backtrack

4. $A \xrightarrow{b} A$

5. backtrack

6. $A \xrightarrow{b} A$

4. 1. opinion

2. sentiment

3. sentiment

4. opinion

5. opinion

6. sentiment

7. sentiment

8. sentiment

9. sentiment

10. opinion

11. opinion

12. sentiment

13. sentiment